Example 2

Solve these simultaneous equations $\begin{aligned} 2x - 5y &= 9 \\ 3x + 2y &= 4 \end{aligned}$

We number the equations ① and ② for convenience.

$$\begin{aligned} 2x - 5y &= 9 \quad ① \\ 3x + 2y &= 4 \quad ② \end{aligned}$$

We now multiply equation ① by 3 and equation ② by 2 to equate the number of x's.

$$\begin{aligned} ① \times 3: \quad 6x - 15y &= 27 \\ ② \times 2: \quad 6x + 4y &= 8 \end{aligned}$$

Subtract:

$$\begin{aligned} -19y &= 19 \\ 19y &= -19 \quad \text{... multiply both sides by } -1 \\ y &= -1 \end{aligned}$$

We now substitute -1 for y in equation ①

$$\begin{aligned} & 2x - 5y = 9 \\ y = -1 \quad \Rightarrow \quad & 2x + 5 = 9 \\ \Rightarrow \quad & 2x = 4 \quad \Rightarrow \quad x = 2 \end{aligned}$$

$\therefore \quad x = 2$ and $y = -1$ are the solutions.

Exercise 12.2

Solve the following simultaneous equations:

1. $x + y = 9$
$x - y = 3$

2. $2x + y = 8$
$3x - y = 2$

3. $2x - y = 4$
$x + y = 5$

4. $3x + y = 7$
$x + y = 5$

5. $4x + y = 17$
$2x + y = 11$

6. $x + 2y = 4$
$x - y = 1$

7. $x + 2y = 7$
$2x + y = 8$

8. $2x - 3y = 1$
$2x + 5y = 9$

9. $2x + 3y = 3$
$x - 4y = 7$

10. $2x - y = -8$
$x + 2y = 6$

11. $4x - y = 10$
$x - y = 1$

12. $4x - y = -9$
$2x - 3y = -7$

13. $3x - y = 3$
$x + 3y = 11$

14. $2x + y = -2$
$x + 3y = 9$

15. $2x - 3y = 14$
$2x - y = 10$

16. $3x + 4y = 5$
$2x - 3y = 9$

17. $3x + y = 5$
$5x - 4y = -3$

18. $2x - y = 12$
$3x + 2y = 11$